IPCON® PLA FC

Food Contact Polylactic Acid Filament

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.03 mm
Net Filament Weight	1 kg ± 0.5%
Filament Length	300 ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C)
Spool Size	Outer Cardboard's diameter: 200 mm; Inner roller's diameter: 90 mm

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.39 g/cm³
Shrinkage Rate*	Specially-defined	0.03% - 0.30%
Surface Hardness	Shore hardness	81D
Saturated Water Absorption Rate	25 °C，55% RH，room air	0.40%
Melt Index	210 °C，2.16 kg	12.0 g/10 min
Melting Temperature	DSC，10 °C/min	153 °C
Vicar Softening Temperature	ISO 306，GB/T 1633	60 °C
Heat Deflection Temperature	ISO 75，0.45 MPa	55 °C

Mechanical Properties*	Testing Methods	Data*
Tensile Strength XY	ISO 527，GB/T 1040	34 MPa
Tensile Strength Z	ISO 527, GB/T 1040	19 MPa
Young's Modulus XY	ISO 527，GB/T 1040	2683 MPa
Young's Modulus Z	ISO 527，GB/T 1040	2458 MPa
Breaking Elongation Rate XY	ISO 527，GB/T 1040	10.5%
Breaking Elongation Rate Z	ISO 527，GB/T 1040	7.1%
Flexural Strength XY	ISO 178，GB/T 9341	64 MPa
Flexural Strength Z	ISO 178，GB/T 9341	40 MPa
Flexural Modulus XY	ISO 178，GB/T 9341	2933 MPa
Flexural Modulus Z	ISO 178，GB/T 9341	2693 MPa
Impact Strength XY	ISO 179，GB/T 1043	29.6 kJ/m²
Impact Strength Z	ISO 179，GB/T 1043	6.1 kJ/m²

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu P1S printer (nozzle temperature: 220 °C, bed temperature: 55 °C, printing speed: XY—270 mm/s and Z—150 mm/s) and IPCON PLA FC (Matte Blue) filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill

pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

1. Most models can be glue-free, food-contact, safe and environmentally friendly, rich in color, high-grade matte and delicate surface, high-speed and easy to print (max speed 300 mm/s), suitable for all mainstream printers, AMS and CFS.
2. Kitchen tools, food containers, pastry molds, children's toys, pet feeders and other items that come into contact with food or mouth.

• Recommended Printing Settings

Drying Settings Before Printing	55 - 60 °C, 6 - 8 h (Blast drying oven)
Printing Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C，≤ 20% RH (Sealed with desiccant)
Compatible Support Material	Itself or support filament for PLA
Compatible Printer Type	Open-frame, enclosed-frame (Suggest to open the front door and remove the upper lad)
AMS or CFS	Compatible
Compatible Nozzle Material	Stainless steel, hardened steel, tungsten carbide, diamond and other materials with very low risk of containing heavy metals (consult the nozzle manufacturer for details)
Compatible Nozzle Size	0.2 / 0.4 / 0.6 / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plate
Plate Surface Preparation	Food contact use: do not use glue or apply food grade glue Non-food contact use: apply liquid glue or PVP solid glue stick
Nozzle Temperature	190 - 230 °C
Bed Temperature	35 - 65 °C
Fan Speed	60% - 100%
Max Print Speed	300 mm/s

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